MULTIPLICATION OF A GROUP SCHEME OVER A FIELD IS AN OPEN MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Lemma morphisms-lemma-scheme-over-field-universally-open (canonical tag 0383). cited at proof lines 960-961. The projection G x_k G -> G is open.

Source reference. Original lemma 0383: sources/stacks-chow/tags.txt, source lines 4222–4222. Cited at proof lines 960-961: the projection pr_0 is open; load-bearing.

Lemma 0.2. *If (G, m) is a group scheme over a field k, then the multiplication map $m : G \times_k G \to G$ is open.*

Proof. The multiplication map is isomorphic to the projection map $\mathrm{pr}_0 : G \times_k G \to G$ because the diagram

$$\begin{array}{ccc} G \times_k G & \xrightarrow[(g,g') \mapsto (m(g,g'),g')]{} & G \times_k G \\ \downarrow{\scriptstyle m} & & \downarrow{\scriptstyle (g,g') \mapsto g} \\ G & \xrightarrow{\mathrm{id}} & G \end{array}$$

is commutative with isomorphisms as horizontal arrows. The projection is open by Morphisms, Lemma 0383. □

Date: stacks-project:groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.